Spectral–Spatial–Language Fusion Network for Hyperspectral, LiDAR, and Text Data Classification

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***Abstract*— The fusion classification of hyperspectral image (HSI) and light detection and ranging (LiDAR) data has gained widespread attention because of its ability to obtain more comprehensive spatial and spectral information. However, the heterogeneous gap between HSI and LiDAR data also adversely affects the classification performance. Despite the excellent performance of traditional multimodal fusion classification models, language information containing much linguistic priori knowledge to enrich visual representations needs to be addressed. Therefore, we design a spectral–spatial–language fusion network (S^2LFNet), which can fuse visual and language features to broaden the semantic space using linguistic priori knowledge commonly shared between spectral features and spatial features. First, we propose a dual-channel cascaded image fusion encoder (DCIFencoder) for visual feature extraction and progressive feature fusion of different levels for HSI and LiDAR data. Then, three aspects of text data are designed to extract linguistic priori knowledge using the text encoder. Finally, contrastive learning is utilized to construct a unified semantic space, and spectral–spatial–language fusion features are obtained for classification tasks. We evaluate the classification performance of the proposed S^2LFNet on three datasets through extensive experiments, and the results show that it outperforms the state-of-the-art fusion classification methods.**

***Index Terms*— Contrastive learning, hyperspectral image (HSI) classification, image classification, light detection and ranging (LiDAR), multimodal.**

I. Introduction

HIGHLY due to the different technical specifications of sensors and actual acquisition conditions in remote sensing images, there is variability among data [1]. For example, a hyperspectral image (HSI) has multiple spectral bands and continuous frequency bands [2]. It carries a large amount of information, which is useful in studying various spectral matching tasks [3]. However, sometimes it is difficult to obtain good classification results based on spectral information alone [4]. Unlike HSI, light detection and ranging (LiDAR) data can be used to record an object's elevation information and acquire high spatial resolution [5]. Therefore, the HSI–LiDAR fusion is a good research direction, and various tasks such as target recognition [6] and classification [7] have emerged. The current mainstream fusion methods can be categorized into feature- and decision-level fusion [8]. Feature-level fusion usually obtains classification results by connecting HSI with features extracted by LiDAR [9]. Decision-level fusion can fuse extracted features and recognized data sources through different fusion strategies according to specific rules [10]. While both feature-level fusion and decision-level fusion techniques are efficient, they differ in the fusion process's stage.

Deep learning methods have become popular in multimodal fusion challenges in recent years [11]. Due to their excellent visual representational skills, these methods achieve extraordinary results [12], [13]. To extract different levels of features in the spectral and spatial dimensions, many branching convolutional neural networks (CNNs) [14], [15] were proposed to capture spectral features and spatial features, respectively. However, the ability to model the sequence correlation of spectral sequences and extract hierarchical spatial features needs to be improved. Therefore, some self-attentive mechanisms [16] and transformers [17] were proposed to make use of both the strong generalization capacity across several domains and the potent modeling capabilities of long-range dependencies [18]. Even though most deep learning models are good at extracting multilevel visual features from HSI and LiDAR, the issue of needing a lot of labeled samples for model training remains [19]. Both HSI and LiDAR data provide challenges in acquiring valuable samples, and the process of gathering labeled data is time-consuming and labor-intensive [20]. In order to compensate for the scarcity of labeled samples, some data enhancement strategies [21] and semi-supervised learning algorithms [22] are used for multimodal fusion classification tasks. Nevertheless, the utilization of data enhancement techniques and the incorporation of unlabeled samples can potentially amplify the disparity in distribution between the two data [23], [24].

To address this issue, the utilization of linguistic prior knowledge, such as the identification of label correlations among classes, can contribute to the enhancement of the

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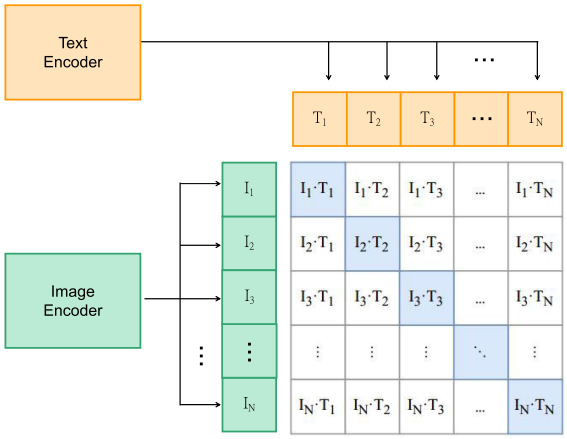


Fig. 1. Commonly used multimodal fusion models; the diagram shows the alignment of the image-text pairs using contrast learning.

extracted visual features' representation. Some image-text foundation models were proposed to capture image-text pairs using contrastive learning. As shown in Fig. 1, unlike simple image multimodal fusion methods, it generates free and prosperous language information by extending the semantic space to improve image classification performance [25]. In practical multimodal classification applications, fully considering the interactions between factors is likewise one of the critical parts of improving classification results [26]. By including text data, these concerns can be effectively tackled by the inclusion of varied descriptions. The extensive land cover classes' linguistic priori knowledge can be obtained from language information [27]. It is more helpful in compensating for the lack of feature sample data. HSI and LiDAR data provide dissimilar modal information as a result of disparities in internal characteristics and external qualities, however they share similar label information [28]. Hence, incorporating language information can serve as a more effective means to merge the complementing characteristics of HSI and LiDAR.

Therefore, we add text data to the two-mode fusion task of the original HSI and LiDAR data, and then we get a three-mode fusion task from the perspective of spectral–spatial–language. The linguistic priori knowledge derived from language information is employed to integrate spectral information with spatial information, resulting in a representation of quality texture features. However, image-text foundation models also have some drawbacks, which can hinder further performance improvement. The following are examples.

1) Most image encoders in image-text models use stacked architectures [29]. However, this blind stacking image encoder does not meet the requirement of extracting sufficient spectra and spatial features for HSI–LiDAR data fusion. Instead, it exacerbates the loss of feature information for both HSI and LiDAR data [30], which limits the improvement of classification results.
2) Although text data is one of the modal information, it is often a single input with a label description. This way may weaken the extraction of multilevel visual features between HSI and LiDAR modal information. In classification tasks with high sample similarity and dense class distribution, a basic label description alone is insufficient [27].

To address the above problems, we propose a three-mode fusion network capable of fusing HSI, LiDAR, and text, called spectral–spatial–language fusion network (S^2LFNet). The image encoder designed for HSI–LiDAR fusion can adequately extract spatial features and spectral features to enhance the representation of fusion visual information. Moreover, the S^2LFNet can extract multilevel linguistic priori knowledge from three different text descriptions and then broaden the semantic space of traditional multimodal fusion classification models. The S^2LFNet is divided into three parts: the image encoder, the text encoder, and the spectral–spatial–language fusion. In the image encoder, a dual-channel cascade image fusion encoder (DCIFencoder) is designed to extract deep spectral features and edge-enhanced spatial features, respectively, and realize progressive incremental multimodal feature-level fusion of HSI and LiDAR data. It solves the problem of feature information loss of HSI and LiDAR data and enhances the model fusion ability. Then, three aspects of text data are designed to extract language features as linguistic priori knowledge using a text encoder, avoiding the weakening of the model by too single-labeled text. Finally, visual features and linguistic priori knowledge are aligned using contrastive learning to construct a unified semantic space, resulting in spectral–spatial–language features for the final classification task. The main contributions of this work are summarized below.

1) To our knowledge, this is the first spectral–spatial–language multimodal fusion network that unifies HSI, LiDAR, and text data, using linguistic priori knowledge as a supervisory signal for contrastive learning to acquire multimodal fusion features.
2) The DCIFencoder has been specifically developed to facilitate the incremental integration of HSI and LiDAR data through a cascade structure. The proposed method acquires deep spectral features through the spectral enhancement channel (SpeEC) and edge enhancement spatial features through the spatial enhancement channel (SpaEC) of the Sobel operator.
3) Three kinds of text data are designed to extract language features with different levels to be aligned with visual features, broadening traditional fusion models' semantic space.

II. Related Work

A. HSI–LiDAR Fusion Classification

Although HSI data have sufficient spectral information, spectral information alone may be detrimental to enhancing classification results in some environments [4]. On the other hand, LiDAR data can supplement HSI data because it has enough spatial information to enhance classification performance. Therefore, using HSI and LiDAR data as a heterogeneous fusion technique has become a hot topic in recent years [31]. Ghamisi et al. [32] were the first to propose a deep learning network for HSI–LiDAR fusion, which utilizes extrapolation profiles (EPs) to achieve feature extraction and then integrate them by feature stacking

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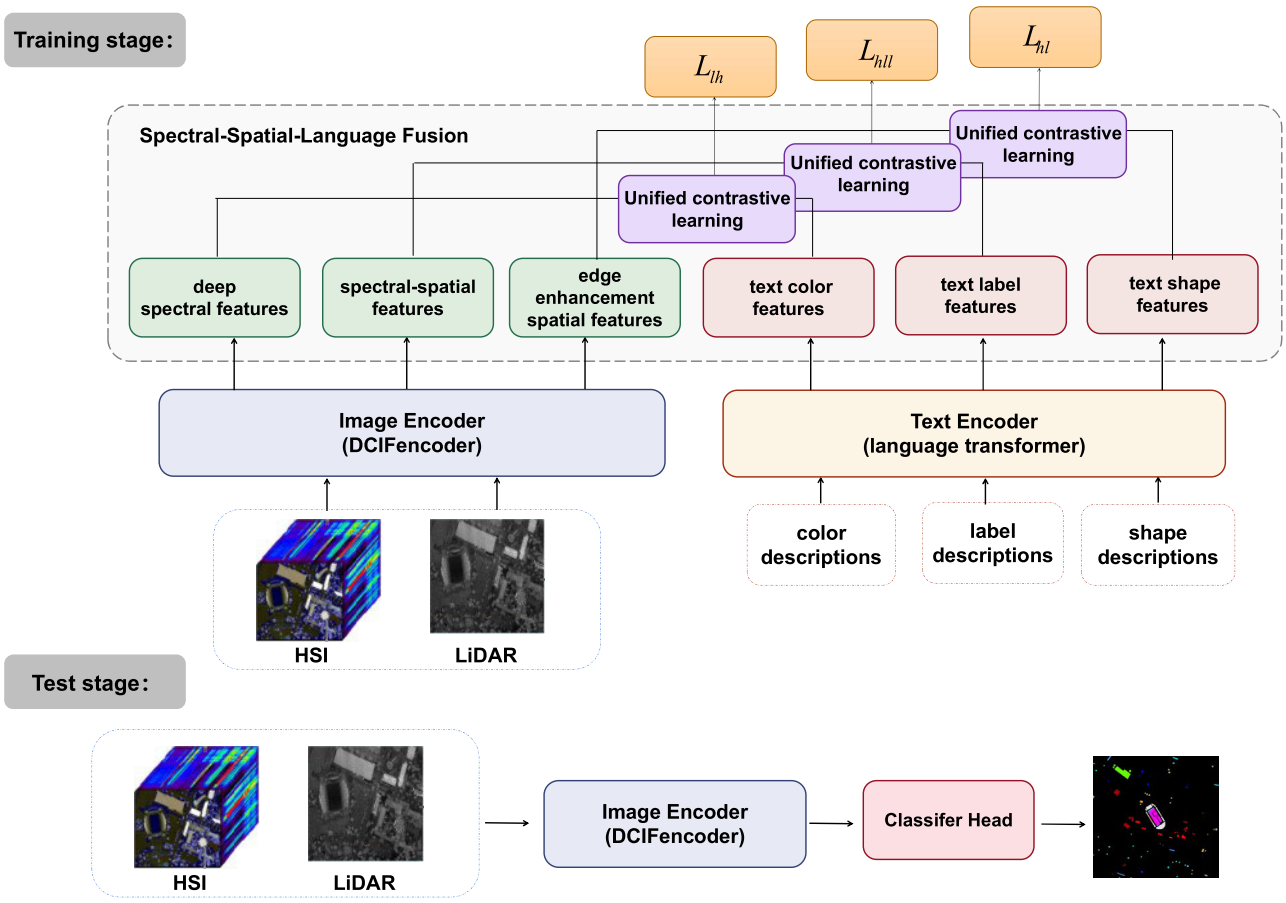


Fig. 2. Flowchart of the proposed S²LFNet. S²LFNet is divided into three parts: an image encoder, a text encoder, and a spectral–spatial–language fusion. The image encoder is designed to extract deep spectral features, spectral–spatial features, and edge enhancement spatial features. The text encoder, a generic language transformer, is used to extract language features as linguistic priori knowledge. Finally, the spectral–spatial–language fusion creates a unified multimodal semantic space.

or graph-based feature fusion to feed them into a deep learning-based classifier for obtaining classification results. Although incorporating deep learning methods significantly improve classification performance, the loss of important feature information may be exacerbated by overly simplistic feature extraction methods and fusion steps. Therefore, scholars have greatly progressed the feature extraction and fusion steps. In feature extraction, Chen et al. [33] proposed a two-branch network to extract spectral features and spatial features separately. The classification results were obtained by integrating the features through a fully connected layer. Other specially designed feature extraction networks include three-branch networks [34], two-tunnel networks [14], and encoder–decoder translation networks [35]. In terms of feature fusion, Feng et al. [36] used squeeze-excitation networks for feature fusion to obtain calibrated features adaptively. There are other effective fusion strategies, such as cross-focus modules [16], reconstruction-based networks [37], combined feature-decision fusion networks [38] and graph fusion networks [39]. Unlike the fusion method utilizing HSI and LiDAR data, Hang et al. [40] proposed an approach that uses paired HSI and LiDAR data for pre-training to extract sufficient information and then transfer these features for HSI classification through fine-tuning.

B. Image–Text Foundation Models

Recently, learning images directly from text data is a rapidly growing field. It can obtain good performance using a more comprehensive range of supervised sources. Existing works can be broadly categorized into two types. The first type aims at learning pre-extracted features from a given image and text encoder using a generalized multimodal fusion layer based on mask-token prediction or image-text matching, e.g., Vilt [41] and Oscar [42]. Their research focuses on improving downstream tasks such as image captioning [43] and visual quizzing [44]. Another type of work focuses on encoding images and text independently to learn transferable visual representations from natural language contrastive learning, e.g., contrastive language image pre-training (CLIP) [45] and CoCa [46]. By putting together pairs of positive samples that are similar and pairs of negative samples that are different, the positive pairs are near together in the projected space, and the opposing pairs are separated from one another. These works investigate the possibility of successfully migrating task-agnostic pre-training in NLP to the visual domain. Such task learning can be achieved by natural language cues for zero-shot migration to many existing datasets. Moreover, the performance of this approach can compete with task-specific supervised models. This article uses contrastive learning for

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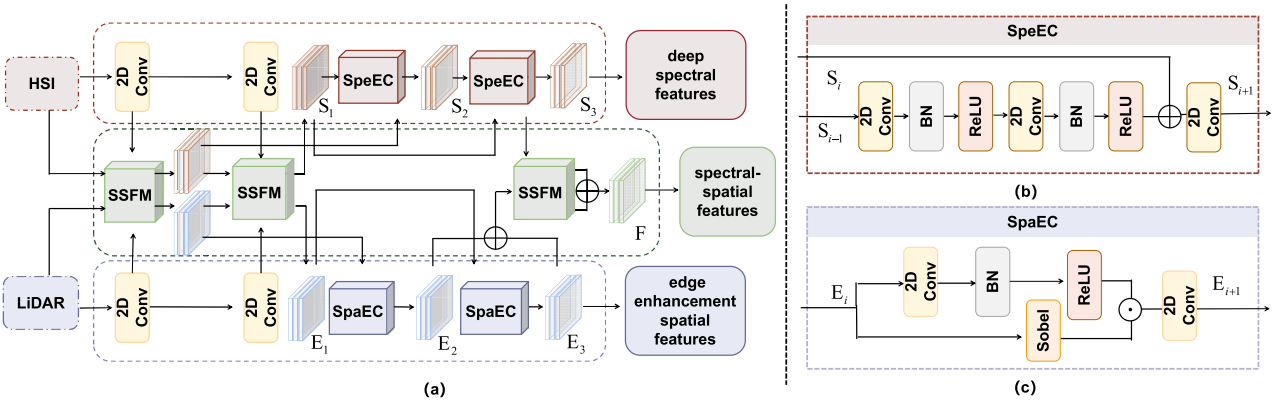


Fig. 3. DCIFencoder. (a) Overall structure. (b) and (c) SpeEC and SpaEC.

S^{2}LFNet to achieve a flexible design and relatively low computational cost.

Although these efforts have exploited the spectral features of HSI data and the spatial features of LiDAR data to the extent possible, there are still bottlenecks in the classification task in practical applications due to the high expense of labeled remote sensing data. It is challenging to produce reliable deep learning models without sufficient labeled samples with just visual feature extractors [47]. To address this problem, we aim to create a three-mode fusion task from the perspective of spectral–spatial–language.

III. Proposed Method

A. Overall Architecture

As shown in Fig. 2, S^{2}LFNet is divided into three parts: image encoder, text encoder, and spectral–spatial–language fusion. The size of the HSI data sample patch is $11 \times 11 \times C_1$ (C_1 denotes the dimensionality of the HSI) and the size of LiDAR data sample patch is $11 \times 11 \times C_2$ (C_2 denotes the dimensionality of the LiDAR data) are selected and fed into the image encoder to extract features, resulting in deep spectral features and edge enhancement spatial features. The deep spectral features and edge enhancement spatial features are fused step by step through the cascaded structure to get the spectral–spatial features. To extract language features and create a shared semantic space, three kinds of text data are designed: label description data, shape description data, and color description data. These three text data are passed to the text encoder to obtain text label features, text spatial features, and text shape features. Finally, multimodal feature matching (specifically, for deep spectral features and text shape features, edge-enhanced spatial features and text spatial features, fusion features and text label features) is performed using contrastive learning to complete three-mode fusion.

B. Image Encoder

In this work, the image encoder is one of the essential components of S^{2}LFNet and significantly impacts the final classification results. However, the majority of image-text models employ stacked image encoders, which fail to fulfill the requirement for complementary information from several modalities in remote sensing tasks. Therefore, we design a DCIFencoder for HSI–LiDAR fusion, as shown in Fig. 3(a). We also verified the DCIFencoder in the experimental section that this structure is more suitable for HSI–LiDAR fusion than a stacked structure. It mainly consists of HSI feature extraction part, LiDAR feature extraction part, spectral–spatial fusion module (SSFM), SpeEC, and SpaEC.

TABLE I
Summary of the Proposed HSI Feature Extraction Part and the LiDAR Feature Extraction Part

HSI feature extraction part			
#	Layer type	Kernel number@size	Output size
	input	——	(11,11,input channel)
1	2D Conv	128@3 × 3	(9,9,128)
2	2D Conv	64@3 × 3	(7,7,64)
LiDAR feature extraction part			
#	Layer type	Kernel number@size	Output size
	input	——	(11,11,1)
1	2D Conv	32@3 × 3	(9,9,32)
2	2D Conv	64@3 × 3	(7,7,64)

The HSI feature extraction parts and LiDAR feature extraction parts consist of two 2-D convolutional layers. The specific input and output parameters and convolutional kernel settings are shown in Table I. Following the process of feature extraction, the features are then passed into the SSFM for the purpose of fusion extraction, the output spectral features are denoted as $\mathbf{S}_1$, and spatial features are denoted as $\mathbf{E}_1$.

The fused spectral features $\mathbf{S}_1$ extracted from SSFM are fed into the SpeEC. Among these, SpeEC is designed as a two-layer structure to obtain the deep spectral features $\mathbf{S}_2$ and $\mathbf{S}_3$ through the spectral enhancement mechanism. Similarly, the fused spatial features $\mathbf{E}_1$ extracted from SSFM are fed into the SpaEC, which is also a two-layer structure designed to obtain the edge enhancement spatial features $\mathbf{E}_2$ and $\mathbf{E}_3$. It is noteworthy that, to achieve a complete fusion of deep features, we use three SSFMs to obtain fusion results step by step and result in spectral–spatial features $\mathbf{F}$.

1) Spectral–Spatial Fusion Module: As shown in Fig. 4, in this study, the specific role of the SSFM is to complete the information interaction of the spectral and spatial features.

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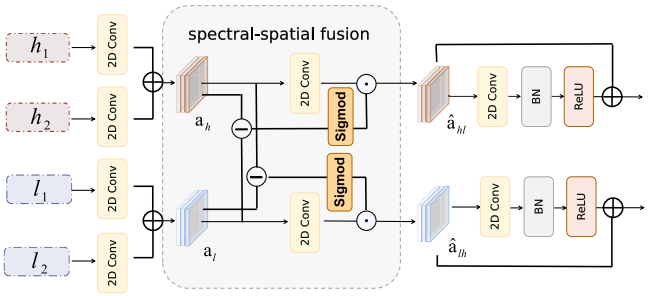


Fig. 4. Model of SSFM in the DCIFencoder.

It can enhance the fusion of the extracted spectral features and spatial features through the cascade structure gradually.

We use two HSI spectral features ($\mathbf{h}_1, \mathbf{h}_2$) and two LiDAR spatial features ($\mathbf{l}_1, \mathbf{l}_2$) extracted from the upper layer as inputs, and all of them are mapped to be of uniform size by a convolutional layer. Spectral features are concatenated as $\mathbf{a}_h$, and spatial features are concatenated as $\mathbf{a}_l$. These features can be represented as

$$\{\mathbf{a}_h, \mathbf{a}_l\} = \left\{C_{\text{spe}}(\mathbf{h}_1) \oplus C_{\text{spe}}(\mathbf{h}_2), C_{\text{spa}}(\mathbf{l}_1) \oplus C_{\text{spa}}(\mathbf{l}_2)\right\} \quad (1)$$

where $C_{\text{spe}}(\cdot)$ and $C_{\text{spa}}(\cdot)$ represent feature extractors for spectral features and spatial features, respectively, and $\oplus$ represents concat operations.

Inputting the obtained $\mathbf{a}_h$ and $\mathbf{a}_l$ into the spectral–spatial fusion structure to obtain the fusion interaction spectral features $\hat{\mathbf{a}}_{hl}$ and the fusion interaction spatial features $\hat{\mathbf{a}}_{hl}$, the process can be expressed as

$$\begin{aligned}\hat{\mathbf{a}}_{hl} &= \mathbf{a}_h + \text{Conv}(\mathbf{a}_h) \odot \text{Sig}(\mathbf{a}_h - \mathbf{a}_l) \\ \hat{\mathbf{a}}_{lh} &= \mathbf{a}_l + \text{Conv}(\mathbf{a}_l) \odot \text{Sig}(\mathbf{a}_l - \mathbf{a}_h)\end{aligned} \quad (2)$$

where $\text{Conv}(\cdot)$ represents the convolution operation, $+$ represents addition operations, $\odot$ represents a dot-multiply operation, and $\text{Sig}(\cdot)$ stands for Sigmod operation.

The $\hat{\mathbf{a}}_{hl}$ and $\hat{\mathbf{a}}_{hl}$ are then convolved separately for a further deep fusion of spectral features and spatial features, and finally, a residual operation is performed to output.

2) Spectral Enhancement Channel: The HSI contains a wealth of spectral information, but it is a challenge to extract and utilize the spectral information to its full potential. In this article, a SpeEC is proposed for the hierarchical extraction of deep spectral features in Fig. 3(b).

It mainly contains two convolution blocks and a connection operation. SpeEC aims to obtain rich spectral features by extracting deep and shallow spectral information, allowing for more adequate spectral features later. The process is shown as follows:

$$\mathbf{S}_{i+1} = \text{Conv}(\text{Conv}(\mathbf{S}_{i-1}) + \mathbf{S}_i) \quad (3)$$

where $\text{Conv}(\cdot)$ represents the convolution operation and $+$ represents the addition operations.

3) Spatial Enhancement Channel: LiDAR, as a 3-D image, contains spatial information that is lacking in HSI. In order to obtain spatial features to match the spectral features, a SpaEC is designed for the hierarchical extraction of spatial features in Fig. 3(c).

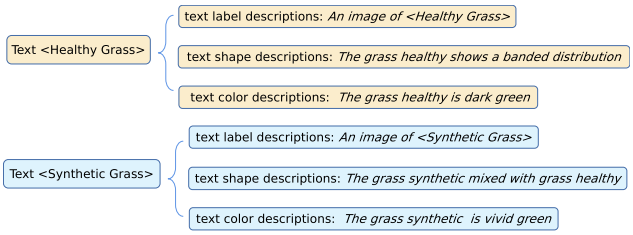


Fig. 5. Text input of the proposed S²LFNet.

It consists of a Sobel operator block and a convolution block. The spatial features enhancement block uses the Sobel operator to perform edge extraction of spatial information, eliminating redundant information and preserving the spatial structural properties. Unlike the SpeEC, the SpaEC does not require the features extracted from the upper layer. Only the spatial information is augmented by a single input. The process is shown below

$$\mathbf{E}_{i+1} = \text{Conv}(\text{Sobel}(\mathbf{E}_i) \odot \text{Sig}(\text{Conv}(\mathbf{E}_i))) \quad (4)$$

where $\text{Conv}(\cdot)$ represents the convolution operation, $\odot$ represents a dot-multiply operations, $\text{Sig}(\cdot)$ stands for the Sigmod operation, and the $\text{Sobel}(\cdot)$ represents the Sobel operator.

C. Text Encoder

The text encoder as an equally pivotal facet of S²LFNet, we leverage the language transformer [48], which is modified by Radford et al. [49]. The modified language transformer uses the transformer-based architecture, which modifies the layer normalization to the input of each sub-block, similar to the pre-activated residual network [50], and adds an additional layer normalization after the final self-attention block. We use the version of the transformer that has three layers and contains 33M parameters. To maintain consistency with CLIP, which is based on image-text contrastive learning, we apply lower-case byte pair encoding (BPE) with a vocabulary size of 49 152 as well [51]. After a layer normalization, language features are projected into the semantic space to match visual features.

According to the linguistic priori knowledge of land cover classes, three different aspects of manually labeled text description will be used to match each land cover class. As shown in Fig. 5, the text label description of each class is manually constructed using the "class name" image as a template [27]. For example, "An image of trees." As for the text shape descriptions, a combination of shape and form is used for the descriptions. For example, "The trees are distributed in ribbons." For the text color descriptions, a combination of colors is used. For example, "The trees are light green."

D. Spectral–Spatial–Language Fusion

In order to perform spectral–spatial–language fusion, the visual features extracted by the image encoder and the language features captured by the text encoder need to construct a unified semantic space. The distance between language features and visual features is reduced by common label construction for contrastive learning. We perform a combination

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of three feature pairs to fuse different modal information: spectral–spatial features and text label features, deep spectral features and text color features, and edge enhancement spatial features and text shape features.

We construct a unified semantic space for the visual features obtained by the image encoder and the language features obtained by the text encoder. Then, using contrastive learning, linguistic priori knowledge is used as a supervisory signal to speak about the alignment of visual and language features by class. A minibatch of all the image-text pairs with identical class labels and one-to-one input data is provided. The loss $\mathcal{L}_{\text{fusion}}$ of spectral–spatial features and label language features can be defined as

$$\mathcal{L}_{\text{fusion}} = -\sum_{i=0}^{N} \frac{1}{|P(i)|} \left(\sum_{k \in P_v(i)} \log \frac{\exp\left(\tau \mathbf{v}_i^T \mathbf{t}_k\right)}{\sum_{j \in A_v(i)} \exp\left(\tau \mathbf{v}_i^T \mathbf{t}_j\right)} + \sum_{k \in P_t(i)} \log \frac{\exp\left(\tau \mathbf{v}_k \mathbf{t}_i^T\right)}{\sum_{j \in A_t(i)} \exp\left(\tau \mathbf{v}_j \mathbf{t}_i^T\right)} \right) \tag{5}$$

where for each visual feature $\mathbf{v}_i$ and language feature $\mathbf{t}_i$ in minibatch, $P_v(i)$ and $A_v(i)$ are the positive and negative sample sets of visual feature, respectively, $P_t(i)$ and $A_t(i)$ are the positive and negative sample sets of linguistic feature, respectively, and $|P_v(i)| = |P_t(i)| = |P(i)|$.

In addition, the loss $\mathcal{L}_{\text{spcectral}}$ of deep spectral features with color language features, and the loss $\mathcal{L}_{\text{spatial}}$ of edge enhancement spatial features with shape language features are both the same as above $\mathcal{L}_{\text{fusion}}$.

Based on the above loss functions, the total loss of S^{2}LFNet is defined as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{fusion}} + \lambda \mathcal{L}_{\text{spcectral}} + \alpha \mathcal{L}_{\text{spatial}} \tag{6}$$

where α and λ are hyperparameters used to control the contribution of language features. The image and text encoder are optimized by the joint loss of the three losses. This enables the linguistic priori knowledge to direct the visual representation's learning process so that features of the same class are clustered, and features of different classes are dispersed. It can help to minimize further the distributional inconsistencies caused by heterogeneous gaps in HSI and LiDAR data and facilitate the fusion of visual data from multiple sources.

IV. Experiments

A. Datasets Description

1) Houston2013 Dataset: The dataset utilized in this article is obtained from the National Airborne Center for Laser Mapping, which serves as a focal point of the 2013 GRSS Data Fusion Contest [52]. The HSI required for this research is gathered by the CASI sensor, which encompasses 144 spectral bands with a spatial resolution of 2.5 m. Complementary LiDAR data, co-registered and sharing an identical resolution, are also available. Within this dataset, a comprehensive collection of 15 classes is represented by 15 029 ground-truth samples. To facilitate experimentation, these samples are segregated into two distinct subsets: a training set containing 2832 pixels and a test set with a more extensive 12 197 pixels. We follow conventional training and test set divisions to ensure consistency with established practices. The specific distribution of training and test samples can be found in Table II.

TABLE II
Land-Cover Classes of the Houston2013 Dataset With the Standard Training and Testing Sets for Each Class

Class NO.	Class Name	Training	Testing
1	Healthy Grass	198	1384
2	Stressed Grass	190	784
3	Synthetic Grass	192	184
4	Tree	188	447
5	Soil	186	694
6	Water	182	439
7	Residential	196	918
8	Commercial	191	2418
9	Road	193	564
10	Highway	191	162
11	Railway	181	1244
12	Parking Lot1	192	330
13	Parking Lot2	184	45
14	Tennis Court	181	39
15	Running Track	187	11
	total	2832	12197

TABLE III
Land-Cover Classes of the Trento Dataset With the Standard Training and Testing Sets for Each Class

Class NO.	Class Name	Training	Testing
1	Apple trees	129	3905
2	Buildings	125	2778
3	Ground	105	374
4	Wood	374	8969
5	Vineyard	184	10317
6	Roads	122	3052
	total	819	29595

2) Trento Dataset: The dataset is acquired from a rural area located to the south of Trento, Italy. [53] The HSI is composed of 63 bands, spanning a wavelength range from 0.42 to 0.99 μm. Spanning dimensions of 166 $\times$ 660 pixels, the dataset maintains a spatial resolution of 1.0 m. Within this dataset, a sum total of 30 214 ground-truth samples is allocated across six distinct classes. A comprehensive breakdown of the distribution of training and test samples for the Trento dataset can be found in Table III.

3) MUUFL Dataset: The dataset was acquired from the University of Southern Mississippi Gulf Coast campus during the month of November 2010 [54], [55]. The HSI comprises a total of 72 spectral bands. However, for the purpose of noise reduction, the initial and concluding four bands are excluded, resulting in 64 bands that are employed for the classification task. The dataset's dimensions encompass 325 $\times$ 220 pixels in total. A comprehensive breakdown of available training and test samples for this dataset is detailed in Table IV.

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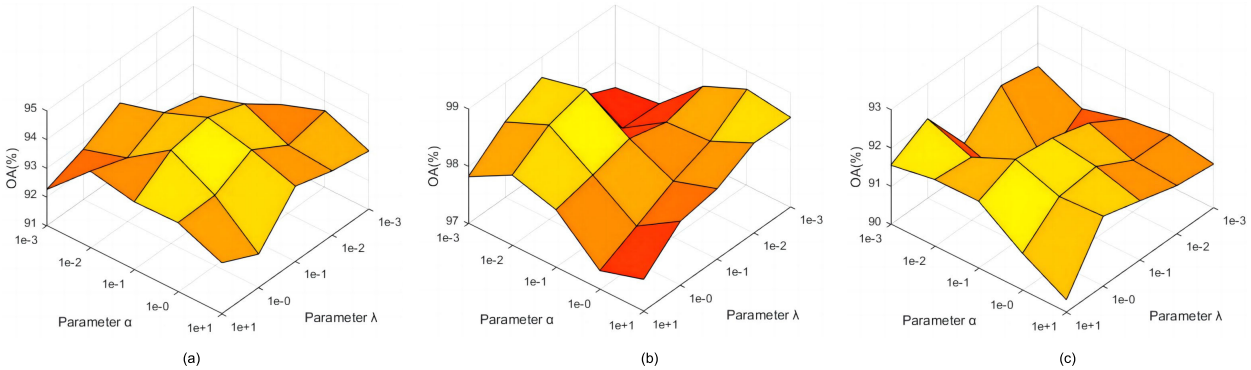


Fig. 6. Parameter tuning of λ and α for the proposed S^{2}LFNet using all the three experimental data. (a) Results on the Houston 2013 dataset. (b) Results on the Trento dataset. (c) Results on the MUUFL dataset.

TABLE IV
LAND-COVER CLASSES OF THE MUUFL UNIVERSITY DATASET WITH THE STANDARD TRAINING AND TESTING SETS FOR EACH CLASS

Class NO.	Class Name	Training	Testing
1	Trees	150	23096
2	Mostly grass	150	4120
3	Mixed ground surface	150	6732
4	Dirt and sand	150	1676
5	Road	150	6537
6	Water	150	316
7	Building shandow	150	2083
8	Building	150	6090
9	Sidewalk	150	1235
10	Yellow curb	150	33
11	Cloth panels	150	119
	total	1650	52037

Model performance assessment encompasses the utilization of three key metrics: overall accuracy (OA), average accuracy (AA), and the κ coefficient. Specifically, OA quantifies the proportion of accurately predicted outcomes by the model against the total count of predictions made on the entire array of test samples. Meanwhile, AA quantifies the precision of predictions within individual labels by determining the ratio of accurate predictions within each label against the overall count within that same label. Additionally, AA is averaged across all labels to provide an aggregate measure. Finally, κ coefficient assesses the level of agreement between predictions and actual values, accounting for chance-level agreements that might arise randomly. This coefficient is essentially the percentage of agreement that surpasses what could be anticipated due to chance alone.

B. Experimental Setup

To validate the validity of S^{2}LFNet, we conduct extensive experiments on three widely used benchmark datasets: 2013 Houston, Trento, and MUUFL. We first compare S^{2}LFNet with the eight state-of-the-art methods. Following that, we carried out thorough ablation tests with the objective of determining the specific influence and efficacy of each element of our strategy.

During the training process, we use the Adam optimizer. The mini-batch size is set to 64, and the learning rate is set to 0.0005. The image patch of 11 $\times$ 11 pixels is randomly cropped from the dataset as training samples.

C. Parameter Tuning

We performed sensitivity analyses of the parameters on the three datasets. The regularization parameters λ and α are regarded as adjustable parameters selected from 1e-3, 1e-2, 1e-1, 1e-0, and 1e+1, respectively.

Fig. 6 shows the OA performance of the three experimental datasets under the variation of the regularization parameters λ and α. The three datasets have different trends in OA, and the suboptimal ranges of α and λ are mainly distributed between 1e-2 and 1e-0. It can be seen that the optimal λ for the Houston 2013 dataset is 1e-1, and the optimal α is 1e-1. The optimal λ for the Trento dataset is 1e-1, and the optimal α is 1e-2. The optimal λ for the MUUFL dataset is 1e-0, and the optimal α is 1e-1. We parameterize each dataset according to its optimal value.

D. Comparison With Representative Baseline Methods

S^{2}LFNet is implemented on the Houston 2013, Trento, and MUUFL datasets. To verify the effectiveness of S^{2}LFNet, we compare it with eight state-of-the-art methods, including FusAtNet [16], EndNet [56], MDL [8], S2ENet [36], CCR-Net [57], MAHiDFNet [58], Sal2RN [20], and LDGnet [27]. In particular, FusAtNet combines the characterization of HSI and LiDAR features through cross-attention and feature embedding by computing specific modalities. In EndNet, a two-branch convolution neural network is utilized for HSI–LiDAR fusion and classification. In MDL, the generalized multimodal deep learning framework is utilized for information fusion. S2ENet simultaneously improves spatial and spectral features characterization through spatial branching networks and spectral branching networks. CCR-Net utilizes cross-modal reconstruction strategies to learn to facilitate fusion of multimodal representations, and uses a dual-channel strategy to efficiently exchange information with each other. MAHiDFNet uses a three-branch strategy to not only extract spatial and spectral information, but also provide

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TABLE V

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS FOR OA, AA, AND κ, AND ACCURACY PER CLASS ON THE DATASET HOUSTON2013. THE BEST CLASSIFICATION METHODS ARE SHOWN IN BOLD

CLASS NO.	FusAtNet	EndNet	MDL	S2ENet	CCRNet	MAHiDFNet	Sal2RN	LDGnet	Ours
1	83.00±0.84	84.24±0.32	83.10±0.41	83.10±0.04	83.10±0.24	83.36±0.22	88.23±0.11	83.10±0.54	**90.46±0.24**
2	97.57±1.32	95.58±1.24	98.40±0.58	98.30±1.23	99.55±0.41	97.25±1.22	**100.00±0.00**	99.25±0.22	95.58±0.04
3	93.22±0.75	99.53±0.21	94.62±2.23	96.36±1.67	98.22±0.11	96.35±0.12	94.58±0.47	94.40±0.25	**100.00±0.00**
4	99.31±0.04	97.92±0.42	97.81±1.32	**99.71±0.13**	95.24±0.27	94.28±0.43	99.53±0.56	97.73±0.22	99.65±0.14
5	99.35±0.25	99.43±0.14	99.32±0.19	**100.00±0.00**	99.81±0.57	99.43±0.52	99.63±0.36	97.16±0.35	97.63±1.38
6	96.50±1.14	95.10±2.21	95.80±0.44	95.80±0.14	98.32±0.46	97.60±0.16	95.80±0.08	96.50±0.17	**99.88±0.34**
7	86.01±0.01	81.53±1.42	93.00±0.36	91.32±0.14	91.88±0.55	93.22±0.96	89.35±0.73	**98.23±0.24**	95.92±0.48
8	86.89±0.98	86.61±0.55	88.32±0.22	93.35±0.74	95.16±1.14	92.32±1.21	90.33±0.34	**95.44±0.87**	91.64±0.35
9	89.52±0.75	84.14±0.14	87.54±0.96	84.99±0.54	84.32±0.24	86.32±0.21	85.65±1.54	66.38±0.64	**90.93±0.77**
10	44.50±2.25	76.45±0.74	94.31±1.12	**95.37±0.14**	94.11±0.47	90.21±0.33	87.25±1.42	87.45±0.24	94.31±0.66
11	95.83±0.24	97.82±1.13	96.11±0.93	98.10±0.75	96.34±0.52	86.34±0.72	96.11±0.58	95.25±0.23	**98.29±0.28**
12	82.13±0.52	**89.63±0.25**	81.17±0.48	84.34±1.24	80.22±0.46	83.44±1.75	85.30±0.24	60.71±1.32	85.11±0.26
13	91.44±0.88	75.79±0.25	92.05±0.47	86.67±1.68	92.33±0.75	85.32±0.15	90.52±0.63	**92.98±0.38**	91.75±1.12
14	99.60±0.87	**100.00±0.00**	98.79±0.77	94.91±1.32	96.36±0.22	**100.00±0.00**	97.80±0.30	97.98±0.24	**100.00±0.00**
15	**100.00±0.00**	98.52±0.22	98.34±1.01	**100.00±0.00**	97.33±0.45	**100.00±0.00**	98.22±0.14	99.31±0.24	**100.00±0.00**
OA(%)	88.35±0.15	90.12±0.41	90.03±0.24	92.62±0.17	93.38±0.24	92.01±0.32	93.64±0.85	89.42±0.76	**94.48±0.13**
AA(%)	90.33±1.24	90.85±0.42	90.79±0.05	93.14±0.80	94.25±0.75	93.32±0.21	94.21±0.21	89.96±0.42	**95.60±0.05**
$\kappa \times 100$	87.36±0.24	89.27±0.44	92.44±0.23	93.07±0.47	92.23±0.45	92.26±0.14	93.10±0.28	88.52±0.12	**94.56±0.22**

TABLE VI

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS FOR OA, AA, AND κ, AND ACCURACY PER CLASS ON THE DATASET TRENTO. THE BEST CLASSIFICATION METHODS ARE SHOWN IN BOLD

CLASS NO.	FusAtNet	EndNet	MDL	S2ENet	CCRNet	MAHiDFNet	Sal2RN	LDGnet	Ours
1	98.93±0.22	99.33±0.75	98.75±1.12	98.10±0.41	99.10±0.04	97.43±0.46	99.13±0.04	98.60±0.75	**99.83±0.11**
2	98.42±0.47	98.59±0.23	98.14±1.14	99.56±0.26	99.43±0.37	**99.81±0.09**	97.83±0.14	98.12±0.32	99.26±0.23
3	97.18±1.75	97.23±0.16	97.36±0.34	98.10±0.36	97.10±0.36	98.99±0.42	93.10±2.36	91.67±0.31	**96.62±0.12**
4	99.47±0.24	99.35±0.47	99.33±0.37	99.63±0.13	**99.75±0.27**	98.23±0.24	98.92±0.48	97.81±0.93	99.43±0.04
5	99.26±0.66	98.86±0.35	98.29±0.75	99.11±0.44	98.83±1.08	97.82±1.13	98.71±1.41	94.66±2.45	**99.48±0.33**
6	**95.23±0.25**	91.49±0.47	88.91±0.35	94.87±0.24	95.29±0.56	94.02±0.75	93.76±0.53	93.27±2.38	94.02±0.36
OA(%)	98.58±0.24	98.24±0.36	97.94±0.31	98.17±0.14	98.64±0.21	98.55±0.34	98.68±0.11	96.65±0.77	**98.82±0.13**
AA(%)	96.81±0.44	96.47±0.27	96.60±0.36	96.38±0.75	96.57±0.41	96.36±0.44	96.62±0.75	94.04±0.26	**97.31±0.14**
$\kappa \times 100$	96.15±0.57	97.67±0.34	96.75±0.32	97.32±0.88	97.26±0.21	98.03±0.14	98.15±0.36	96.32±0.83	**98.23±0.43**

elevation features and further facilitate multimodal information interaction using hierarchical fusion. Sal2RN uses a three-channel spatial–spectral saliency enhancement network and constructs a statistically based feature weighting combination module for multilevel fusion of features. LDGnet is a network that can combine text data for semantic space alignment using deep residual 3-D CNN as an image encoder. However, it is only for the classification of HSI data. In order to make LDGnet cooperate with the multimodal fusion task, referring to the early fusion approach in MDL [8], we combine the HSI and LiDAR data and input them together to LDGnet. All comparison methods were worked with both HSI and LiDAR data as inputs and used preset default parameters.

Table V shows the classification results on the Houston 2013 dataset. S^{2}LFNet achieves the best performance regarding OA, AA, and κ coefficients. This shows that our S^{2}LFNet, because of the effective cooperation of the DCIFencoder and the text data, can obtain better classification results. In particular, S^{2}LFNet is able to obtain excellent classification performance of the combined linguistic priori knowledge with visual features on analogs. A classification chart is shown in Fig. 7. It can be seen that in the absence of text data, some Highway areas are prone to be incorrectly classified as railway, which is relatively similar in type and close in distribution. In contrast, S^{2}LFNet performs better through spectral–spatial–language multimodal fusion.

Table VI shows the classification results of different methods for the Trento dataset. The classification graph is shown in Fig. 8. It can be seen that although the dataset has a small number of classes, the sample size is extremely heterogeneous. For example, the label "Vineyard" has a small number of training samples but a large number of test samples. However,

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TABLE VII

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS FOR OA, AA, AND κ, AND ACCURACY PER CLASS ON THE DATASET MUUFL. THE BEST CLASSIFICATION METHODS ARE SHOWN IN BOLD

CLASS NO.	FusAtNet	EndNet	MDL	S2ENet	CCRNet	MAHiDFNet	Sal2RN	LDGnet	Ours
1	95.2±0.28	96.86±1.32	90.95±0.24	94.95±0.81	93.66±0.28	98.16±0.18	**98.77±0.12**	94.66±0.14	96.86±0.42
2	82.27±1.14	72.09±1.35	**84.54±0.41**	81.64±0.32	81.29±1.47	77.19 ±1.21	80.22±0.58	83.75±0.34	82.27±0.46
3	83.54±0.67	80.24±0.27	83.01±0.34	90.55±0.88	86.29±0.24	**91.64±1.13**	90.72±0.08	72.93±0.44	82.42±0.68
4	87.99±0.72	73.67±0.86	96.42±1.15	83.02±2.17	95.71±1.22	97.54±0.75	65.35±0.27	**94.45±0.15**	94.46±1.47
5	93.33±0.25	96.56±0.36	90.44±2.17	94.50±0.65	93.98±0.44	94.33±1.32	95.95±0.82	94.55±1.25	**96.84±0.96**
6	87.76±0.05	64.89±0.36	92.62±0.75	94.68±0.58	90.06±1.46	98.13±0.48	90.75±1.42	91.34±1.13	**98.68±0.36**
7	76.50±0.55	66.52±1.62	92.75±0.31	88.29±1.85	**93.88±0.17**	82.95±0.07	92.82±0.35	79.79±0.55	92.75±0.42
8	97.16±0.68	95.41±1.24	96.70±1.38	93.42±0.75	90.95±2.24	**97.92±0.56**	92.82±0.25	94.79±0.95	95.84±0.84
9	65.43±0.87	60.45±0.46	77.85±1.75	**88.50±0.97**	85.34±0.56	79.26±0.41	65.34±1.24	74.86±0.22	80.72±0.43
10	18.62±5.56	47.03±3.75	58.36±2.72	62.35±1.46	88.33±0.58	87.88±0.69	88.36±0.35	84.36±1.85	**89.16±0.36**
11	73.34±1.38	83.49±0.96	73.34±1.42	80.18±0.81	94.63±1.42	91.77±0.24	93.36±0.86	94.08±0.43	**95.89±0.57**
OA(%)	91.77±0.85	87.75±0.58	90.27±0.36	91.68±0.87	92.08±0.37	92.13±0.29	90.95±0.24	89.74±0.52	**92.17±0.36**
AA(%)	89.74±0.32	86.11±0.75	88.88±0.33	90.02±0.28	90.93±0.39	90.34±0.17	91.05±0.17	87.64±0.35	**91.53±0.14**
$\kappa \times 100$	89.28±0.77	84.06±0.36	87.26±0.44	89.15±0.38	90.19±0.46	90.75±0.63	90.44±0.54	86.75±0.63	**91.13±0.32**

our S^{2}LFNet does a good job of classifying it. This is because adding text data allows the model to learn rich language features and fine-grained features as a basis for classification despite the limited training samples.

The classification results of the different methods on the MUUFL dataset are given in Table VII. Our method obtains the best performance. There is some interference from the shaded parts in this dataset. However, because of the dual assistance like shape language features and color language features, S^{2}LFNet still achieves good performance on the challenging dataset. The classification diagram of the S^{2}LFNet proposed in this article is shown in Fig. 9. The S^{2}LFNet has the most straightforward class map.

By leveraging linguistic priori information, our method demonstrates superior performance. However, in this study, the potential of incorporating text information into the image-text adaptive learning method may not have been fully utilized. The limited size and source of the text data, as well as our reliance on manually retrieved rather than computationally generated text information, contribute to this situation. Hence, there is still potential for enhancing the benefits yielded by the outcomes.

In addition, we find that LDGnet, a joint visual-linguistic network with simple modifications, performs poorly. This is because the image encoder part of the LDGnet network label is a simple stacked CNN. It does not meet the feature extraction and fusion needs of HSI and LiDAR. This shows that the modification of the image encoder DCIFencoder by our S^{2}LFNet is essential and further illustrates the effectiveness of our approach.

E. Number of Training Samples

Image-text's contrastive learning approach has also achieved better results when dealing with classification tasks with a small number of training samples. We tested all the contrast methods with different numbers of samples. The results are shown in Fig. 10. On all three datasets, S^{2}LFNet always shows good classification performance when using smaller training samples.

TABLE VIII

ABLATION ANALYSIS OF THE PROPOSED S^{2}LFNET WITH A COMBINATION OF DIFFERENT TEXT INPUT ON THE HOUSTON 2013 DATASET

Text data input	Matrix		
	OA(%)	AA(%)	$\kappa \times 100$
w/o Text data	93.30	94.02	92.39
label descriptions	93.56	94.53	93.24
shape descriptions	93.77	94.84	93.49
color descriptions	93.62	94.41	94.02
w/o label descriptions	94.25	94.39	94.32
w/o shape descriptions	94.10	95.07	94.17
w/o color descriptions	94.06	95.10	94.28
three Text data	**94.48**	**95.60**	**94.56**

F. Ablation Experiments

To evaluate the efficiency of the various components of S^{2}LFNet, a series of ablation studies are carried out. The specific experimental results are analyzed in the following ways.

1) Validity of Text Data: We conducted experiments on the three text inputs used in S^{2}LFNet to demonstrate the necessity of their joint experiments. As shown in Table VIII, the classification performance of S^{2}LFNet can be seen clearly that the complete S^{2}LFNet performs the best and the only image encoder part is the worst. Among them, using one alone and jointly making text data with two kinds of text data have different degrees of improvement in performance

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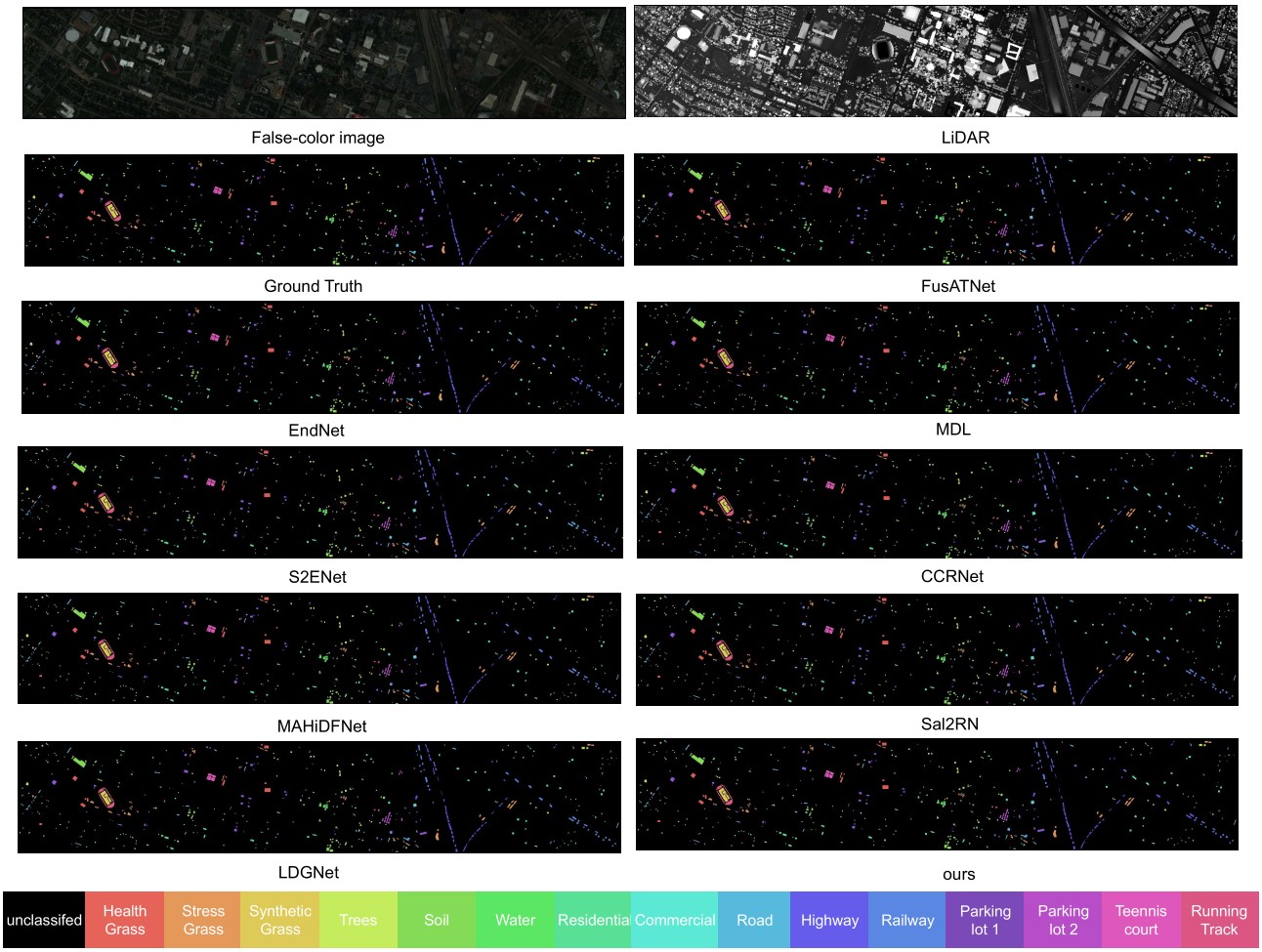


Fig. 7. Classification maps obtained by different models on the Houston 2013 dataset.

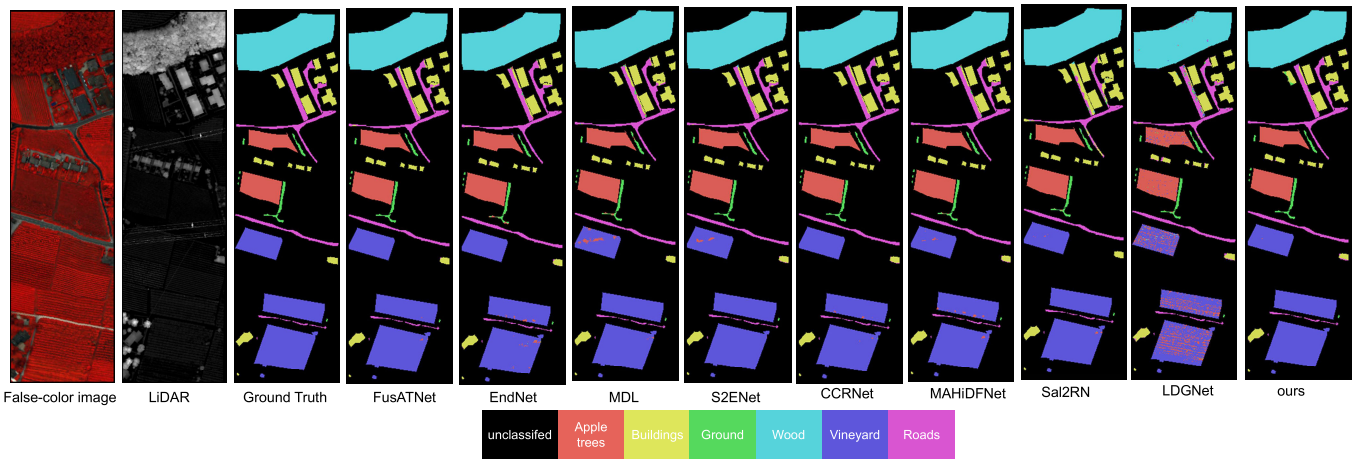


Fig. 8. Classification maps obtained by different models on the Trendo dataset.

compared to not using text data at all. This suggests that label, shape, and color descriptions are all helpful in guiding multimodal feature extraction for the model. This is because the involvement of text data as a linguistic a priori knowledge provides better access to complete text information fusion features and further facilitates information complementarity between HSI and LiDAR data.

2) Validity of Image Data: Considering the effects of different modal data on the classification performance of the model, we compared the differences between the three input modalities: single HSI data, single LiDAR data, and HSI+LiDAR. The experimental results are shown in Table IX. By comparing the OA values of the three datasets, it is verified that the joint input approach of the two types of data is able to obtain more

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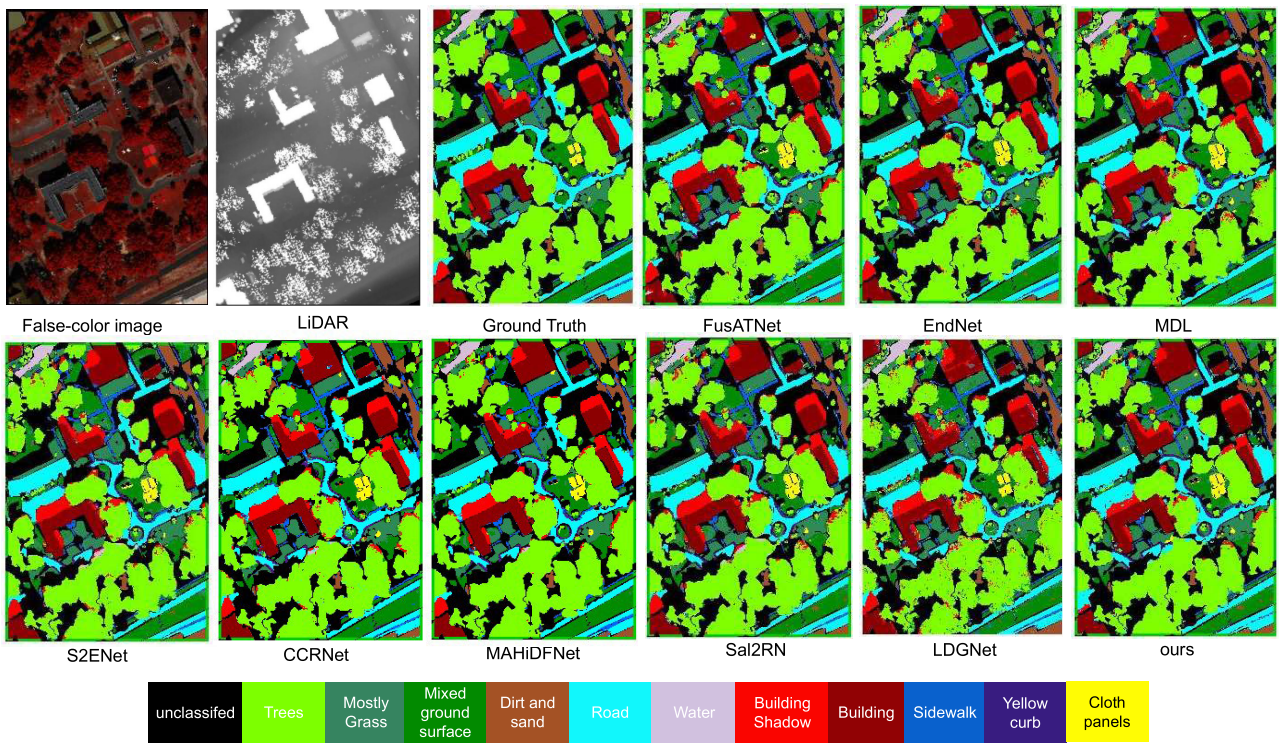


Fig. 9. Classification maps obtained by different models on the MUUFL dataset.

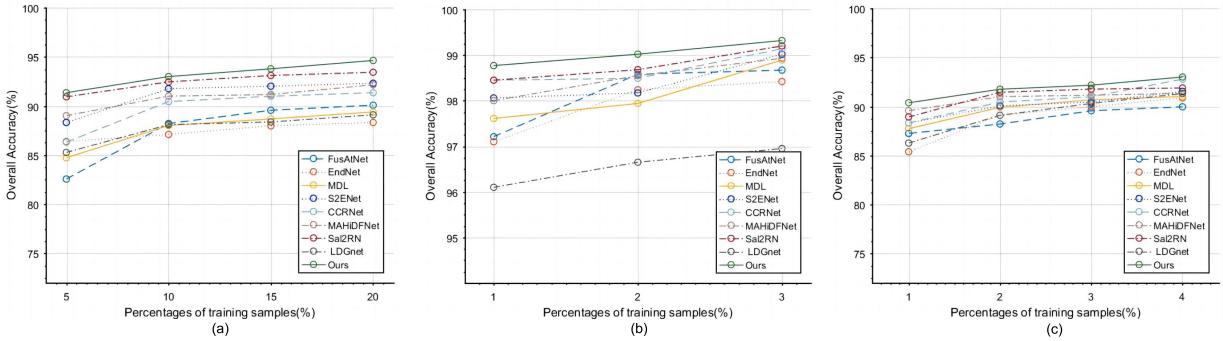


Fig. 10. OA of different models with different training samples percentages. (a) Results on the Houston 2013 dataset. (b) Results on the Trento dataset. (c) Results on the MUUFL dataset.

TABLE IX

OA FOR THE THREE DATASETS WITH DIFFERENT MODAL DATA INPUTS

Image data input	Matrix		
	Houston 2013	Trento	MUUFL
HSI	93.87	98.24	91.35
LiDAR	65.36	91.33	47.39
HSI+LiDAR	**94.48**	**98.82**	**92.17**

TABLE X

ABLATION ANALYSIS OF THE PROPOSED DCIFENCODER WITH A COMBINATION OF DIFFERENT MODULES ON THE HOUSTON 2013 DATASET

No.	Model			Matrix		
	SSFM	cascaded structrue	SpeEC&SpaEC	OA(%)	AA(%)	$\kappa \times 100$
1	✘	✘	✘	89.02	90.86	88.86
2	✘	✘	✓	92.05	93.32	91.19
3	✓	✘	✘	91.55	92.77	90.99
4	✓	✓	✘	91.73	93.08	91.03
5	✓	✓	✓	**93.30**	**94.02**	**0.9239**

adequate spatial–spectral information and ultimately improve the classification performance. It also reflects that the designed two-branch fusion strategy helps to improve the fusion of multimodal information.

3) Validity of the Image Encoder: In this work, the image encoder plays a prominent role in S^{2}LFNet because of the task of multimodal fusion of remote sensing images. However, the simple stacked structure makes it challenging to meet the requirements of feature extraction with a fusion of HSI and LiDAR data. Therefore, we design a DCIFencoder to solve the extraction of spectral features, spatial features, and complete

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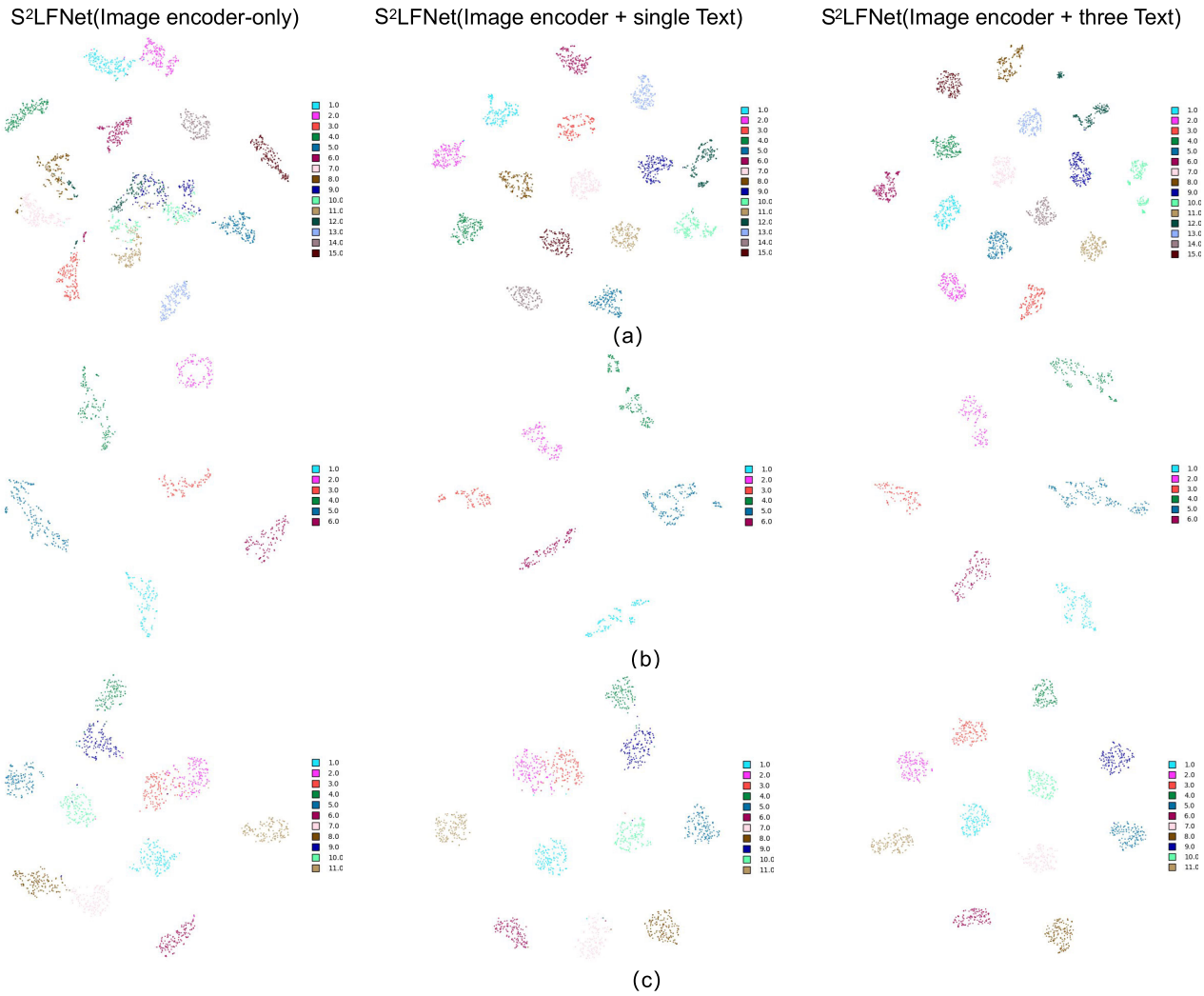


Fig. 11. Feature visualizations on different datasets. (a) Results on the Houston 2013 dataset. (b) Results on the Trento dataset. (c) Results on the MUUFL dataset.

fusion features. We likewise conduct ablation experiments on the single image encoder of S^{2}LFNet to demonstrate the effectiveness of each module and edge extraction operators in the image encoder.

1) *Individual Modules in DCIFencoder:* As shown in Table X, we analyze the role of SSFM, cascaded structure, and SpeEC and SpaEC and whether they have an effect on the DCIFencoder. The classification results are inferior when using only a stacked 3-D CNN network (without SSFM, cascade structure, and SpeEC and SpaEC) as a feature extractor with simple connected feature fusion. However, with the addition of SSFM, the classification performance is significantly and dramatically improved. This is because excessive feature differences exacerbate information loss. At the same time, feature fusion operations can obtain the interaction of spatial and spectral fused information for classification. The addition of the SpeEC and SpaEC dual-channel further improves the classification results. This is because the dual-channel enhances the spatial edge information, deep spectral features and edge enhancement spatial features can realize fuller complementarity. Meanwhile, the rich spectral–spatial fusion features are more conducive to the classification. Finally, through the addition of a cascaded structure, the multimodal information realizes the interaction between deep and shallow layers and further enhances the fusion classification effect by obtaining abundant fusion features at the language level. Finally, the best classification results are obtained under the joint effect of the three.

2) *Sobel Operator in SpaEC:* As shown in Table XI, we further validate the effect of different edge extraction operators and finally prove the effectiveness of the Sobel operator. The selected Roberts operator and Prewitt

TABLE XI
OA OF DIFFERENT EDGE EXTRACTION OPERATORS ON THREE DATASETS

Edge Extraction Operators	Matrix		
	Houston 2013	Trento	MUUFL
Roberts Operator	94.31	98.42	91.56
Prewitt Operator	94.42	98.68	91.94
Sobel Operator	**94.48**	**98.82**	**92.17**

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operator are the same as the Sobel operator, which is one of the commonly used edge extraction methods. The Roberts operator is not very accurate in detecting edges based on local difference computation. As a result, it leads to a loss of spatial information for areas with rich class distributions and dense regions in LiDAR. Consequently, it achieves the lowest OA score. The Prewitt operator employs the discrepancies produced by the grayscale values of a particular area to achieve edge detection, and its horizontal and vertical effectiveness surpasses that of the Roberts operator. It performs well on dense distributions in LiDAR, but still has difficulty in edge extraction on some irregular classes. Thus, it outperforms the Roberts operator in terms of effectiveness, second only to the Sobel operator. Moreover, the Sobel operator combines Gaussian smoothing and differential derivation with the added concept of weights, which results in a sharper image with more prominent edge contours. The efficient edge extraction task adapted to LiDAR is able to obtain more complete spatial information, which further facilitates the fusion of spatial–spectral information. Therefore, among all the edge extraction operators, the Sobel operator has outstanding advantages.

G. Visual Evaluation

To further demonstrate the effectiveness of the text encoder and image encoder, we visualize the three features as shown in Fig. 11. In the 2013 Houston, Trento, and MUUFL datasets, we find that in full S²LFNet, the features with the same label are close to each other, while features with different label are far away from each other. This is due to the fact that the positive and negative sample pairs of image-text constructed by contrastive learning make the positive sample pairs closer and the negative sample pairs farther apart in a uniform feature space.

V. Conclusion

A S²LFNet with three-mode fusion of HSI, LiDAR, and text is proposed. A text representation with linguistic priori knowledge is used to guide the fusion of visual features extracted from HSI and LiDAR data. Specifically, the feature extraction and feature fusion parts for HSI and LiDAR data use a DCIFencoder to obtain visual features. The language features extracted by the text encoder using three text inputs are aligned with the visual features. To facilitate further the fusion of HSI and LiDAR data using contrastive learning, S²LFNet achieves state-of-the-art HSI-LiDAR fusion classification results.

Nevertheless, currently, the acquisition of text data in remote sensing is primarily done manually, with restricted sources and quantities. These constraints can impede the progress of the image-text model in remote sensing. In the future, we will explore more extraction techniques for textual data, investigate the integration of HSI, LiDAR, and text to enhance its applicability in remote sensing classification tasks, and endeavor to develop a streamlined visual language model to minimize network complexity.

References

[1] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[2] Y. Zhang, W. Li, W. Sun, R. Tao, and Q. Du, "Single-source domain expansion network for cross-scene hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1498–1512, 2023.

[3] D. Hong et al., "Interpretable hyperspectral artificial intelligence: When nonconvex modeling meets hyperspectral remote sensing," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 2, pp. 52–87, Jun. 2021.

[4] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[5] B. Chen et al., "Using HSI color space to improve the multispectral LiDAR classification error caused by measurement geometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3567–3579, Apr. 2021.

[6] A. V. Kanaev, B. J. Daniel, J. G. Neumann, A. M. Kim, and K. R. Lee, "Object level HSI-LiDAR data fusion for automated detection of difficult targets," *Opt. Exp.*, vol. 19, no. 21, pp. 20916–20929, 2011.

[7] S. K. Roy, A. Deria, D. Hong, M. Ahmad, A. Plaza, and J. Chanussot, "Hyperspectral and LiDAR data classification using joint CNNs and morphological feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[8] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

[9] C. Demirkesen, M. Teke, and U. Sakarya, "Hyperspectral images and LiDAR based DEM fusion: A multi-modal landuse classification strategy," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 2942–2945.

[10] Y. Zhang, H. L. Yang, S. Prasad, E. Pasolli, J. Jung, and M. Crawford, "Ensemble multiple kernel active learning for classification of multisource remote sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 2, pp. 845–858, Feb. 2015.

[11] L. Tang, Y. Deng, Y. Ma, J. Huang, and J. Ma, "SuperFusion: A versatile image registration and fusion network with semantic awareness," *IEEE/CAA J. Autom. Sinica*, vol. 9, no. 12, pp. 2121–2137, Dec. 2022.

[12] J. Ma, L. Tang, F. Fan, J. Huang, X. Mei, and Y. Ma, "SwinFusion: Cross-domain long-range learning for general image fusion via Swin transformer," *IEEE/CAA J. Autom. Sinica*, vol. 9, no. 7, pp. 1200–1217, Jul. 2022.

[13] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–14, 2021.

[14] X. Zhao et al., "Joint classification of hyperspectral and LiDAR data using hierarchical random walk and deep CNN architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7355–7370, Oct. 2020.

[15] M. Zhang, W. Li, Q. Du, L. Gao, and B. Zhang, "Feature extraction for classification of hyperspectral and LiDAR data using patch-to-patch CNN," *IEEE Trans. Cybern.*, vol. 50, no. 1, pp. 100–111, Jan. 2020.

[16] S. Mohla, S. Pande, B. Banerjee, and S. Chaudhuri, "FusAtNet: Dual attention based SpectroSpatial multimodal fusion network for hyperspectral and LiDAR classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 416–425.

[17] X. Zhao et al., "Fractional Fourier image transformer for multimodal remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–13, 2022.

[18] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[19] M. Zhang, X. Zhao, W. Li, Y. Zhang, R. Tao, and Q. Du, "Cross-scene joint classification of multisource data with multilevel domain adaption network," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–13, 2023.

[20] J. Li, Y. Liu, R. Song, Y. Li, K. Han, and Q. Du, "Sal2RN: A spatial–spectral salient reinforcement network for hyperspectral and LiDAR data fusion classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[21] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[22] J. Li, Y. Ma, R. Song, B. Xi, D. Hong, and Q. Du, "A triplet semisupervised deep network for fusion classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[23] Z.-H. Zhou, "A brief introduction to weakly supervised learning," *Nat. Sci. Rev.*, vol. 5, no. 1, pp. 44–53, Jan. 2018.

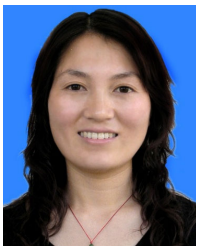
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[24] H. Lee, S. J. Hwang, and J. Shin, "Rethinking data augmentation: Self-supervision and self-distillation," *ArXiv*, vol. abs/1910.05872, 2019.
[25] J. Yang et al., "Unified contrastive learning in image-text-label space," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 19141–19151.
[26] B. Zhou, J. Liu, S. Cui, and Y. Zhao, "Large-scale traffic congestion prediction based on multimodal fusion and representation mapping," 2022, *arXiv:2208.11061*.
[27] Y. Zhang, M. Zhang, W. Li, S. Wang, and R. Tao, "Language-aware domain generalization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.
[28] J. Li et al., "Deep learning in multimodal remote sensing data fusion: A comprehensive review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102926.
[29] M. Tao, H. Tang, F. Wu, X. Jing, B.-K. Bao, and C. Xu, "DF-GAN: A simple and effective baseline for text-to-image synthesis," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16494–16504.
[30] M. E. Paoletti, J. M. Haut, S. K. Roy, and E. M. T. Hendrix, "Rotation equivariant convolutional neural networks for hyperspectral image classification," *IEEE Access*, vol. 8, pp. 179575–179591, 2020.
[31] J. Zhang and X. Lin, "Advances in fusion of optical imagery and LiDAR point cloud applied to photogrammetry and remote sensing," *Int. J. Image Data Fusion*, vol. 8, no. 1, pp. 1–31, Jan. 2017.
[32] P. Ghamisi, B. Höfle, and X. X. Zhu, "Hyperspectral and LiDAR data fusion using extinction profiles and deep convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 6, pp. 3011–3024, Jun. 2017.
[33] Y. Chen, C. Li, P. Ghamisi, X. Jia, and Y. Gu, "Deep fusion of remote sensing data for accurate classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1253–1257, Aug. 2017.
[34] H. Li, P. Ghamisi, U. Soergel, and X. Zhu, "Hyperspectral and LiDAR fusion using deep three-stream convolutional neural networks," *Remote Sens.*, vol. 10, no. 10, p. 1649, Oct. 2018.
[35] Y. Zheng, J. Li, Y. Li, J. Guo, X. Wu, and J. Chanussot, "Hyperspectral pansharpening using deep prior and dual attention residual network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8059–8076, Nov. 2020.
[36] S. Fang, K. Li, and Z. Li, "S^2: Spatial–spectral cross-modal enhancement network for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[37] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[38] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.
[39] X. Du, X. Zheng, X. Lu, and A. A. Doudkin, "Multisource remote sensing data classification with graph fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10062–10072, Dec. 2021.
[40] P. Guan and E. Y. Lam, "Cross-domain contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.
[41] W. Kim, B. Son, and I. Kim, "ViLT: Vision-and-language transformer without convolution or region supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 5583–5594.
[42] X. Li et al., "Oscar: Object-semantics aligned pre-training for vision-language tasks," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*, Glasgow, U.K. Cham, Switzerland: Springer, Aug. 2020, pp. 121–137.
[43] J. Donahue et al., "DeCAF: A deep convolutional activation feature for generic visual recognition," in *Proc. Int. Conf. Mach. Learn.*, 2014, pp. 647–655.
[44] D. A. Hudson and C. D. Manning, "GQA: A new dataset for real-world visual reasoning and compositional question answering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6693–6702.
[45] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.
[46] J. Yu, Z. Wang, V. Vasudevan, L. Yeung, M. Seyedhosseini, and Y. Wu, "CoCa: Contrastive captioners are image-text foundation models," 2022, *arXiv:2205.01917*.
[47] S. Zhang, Z. Chen, D. Wang, and Z. J. Wang, "Cross-domain few-shot contrastive learning for hyperspectral images classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[48] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.
[49] A. Radford et al., "Language models are unsupervised multitask learners," *OpenAI Blog*, vol. 1, no. 8, p. 9, 2019.
[50] K. He, X. Zhang, S. Ren, and J. Sun, "Identity mappings in deep residual networks," in *Proc. 14th Eur. Conf. Comput. Vis. (ECCV)*, Amsterdam, The Netherlands. Cham, Switzerland: Springer, Oct. 2016, pp. 630–645.
[51] R. Sennrich, B. Haddow, and A. Birch, "Neural machine translation of rare words with subword units," 2015, *arXiv:1508.07909*.
[52] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.
[53] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2020.3017414.
[54] X. Du and A. Zare, "Technical report: Scene label ground truth map for MUUFL Gulfport data set," Univ. Florida, Gainesville, FL, USA, Tech. Rep. 20170417, 2017.
[55] P. Gader, A. Zare, R. Close, J. Aitken, and G. Tuell, "MUUFL Gulfport hyperspectral and LiDAR airborne data set," Univ. Florida, Gainesville, FL, USA, Tech. Rep. REP-2013-570, 2013.
[56] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.
[57] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.
[58] X. Wang, Y. Feng, R. Song, Z. Mu, and C. Song, "Multi-attentive hierarchical dense fusion net for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 82, pp. 1–18, Jun. 2022.



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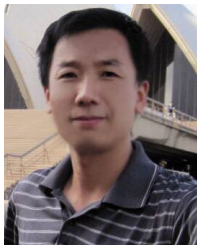
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